
Comparative Study of WGAN-GP and Mini-DDPM for Data Augmentation in Pneumonia Diagnosis from Chest X-Rays

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Abstract

Class imbalance is a major challenge in automated pneumonia diagnosis from chest X-rays, where positive samples are often much fewer than normal cases. This work investigates lightweight generative augmentation under limited computational resources by comparing WGAN-GP and Mini-DDPM on the Kaggle Chest X-Ray Images (Pneumonia) dataset. Both models are trained on pneumonia images only at 64×64 resolution and evaluated from both generative and downstream diagnostic perspectives. Mini-DDPM achieves a lower FID than WGAN-GP (0.0391 vs. 0.0606) and produces visually cleaner samples. More importantly, replacing half of the real pneumonia training images with synthetic ones does not degrade downstream ResNet-18 classification performance in either the ImageNet-pretrained or training-from-scratch setting. These findings suggest that lightweight generative models can provide practically useful medical image augmentation under constrained hardware, with Mini-DDPM offering the best overall balance in our experiments.

1 Introduction

Automated pneumonia diagnosis from chest X-rays is strongly affected by class imbalance, where positive cases are often much fewer than normal controls [1; 2]. This bias pushes classifiers toward the majority class and increases the risk of missed diagnoses [3]. Traditional augmentation methods such as flipping or rotation improve robustness but do not create new pathological variations. Generative augmentation offers a more meaningful alternative, yet many high-performing models are too expensive for standard clinical hardware.

In this work, we study lightweight generative augmentation for pneumonia chest X-rays under computational constraints by comparing WGAN-GP and Mini-DDPM in a unified pipeline. We focus on low-resolution grayscale synthesis and evaluate the generated images not only by visual quality and FID, but also by their effect on downstream pneumonia classification.

2 Related Work

GAN-based augmentation has been widely studied in medical imaging. Prior work shows that synthetic images can improve classifier performance, including in liver lesion classification and chest X-ray analysis [4; 5]. However, standard GANs often suffer from instability and mode collapse, which motivated WGAN-GP as a more stable adversarial framework [6].

Diffusion models generally provide better image quality and mode coverage than GANs, but their computational cost is higher [7]. On the Kaggle Chest X-Ray Images (Pneumonia) dataset, most

previous work has emphasized conventional augmentation and transfer learning [8], while generative augmentation remains less explored. Prior studies suggest that synthetic data can help classification, but also show that better visual quality does not always translate into better downstream performance [9; 10]. This motivates a controlled comparison between lightweight GAN- and diffusion-based augmentation under the same downstream pipeline.

3 Methodology

We compare two lightweight generators, WGAN-GP and Mini-DDPM, for synthesizing pneumonia chest X-ray images. The main question is whether these synthetic images preserve diagnostically useful pneumonia patterns. This is evaluated later through a downstream ResNet-18 classification task (Section 4.3). Here we describe the architectures and training objectives of the two generators.

3.1 WGAN-GP

3.1.1 Model Architecture

Figure 1 shows the WGAN-GP architecture. The generator G maps latent noise to synthetic chest X-rays, and the critic D estimates the Wasserstein distance between real and generated distributions.

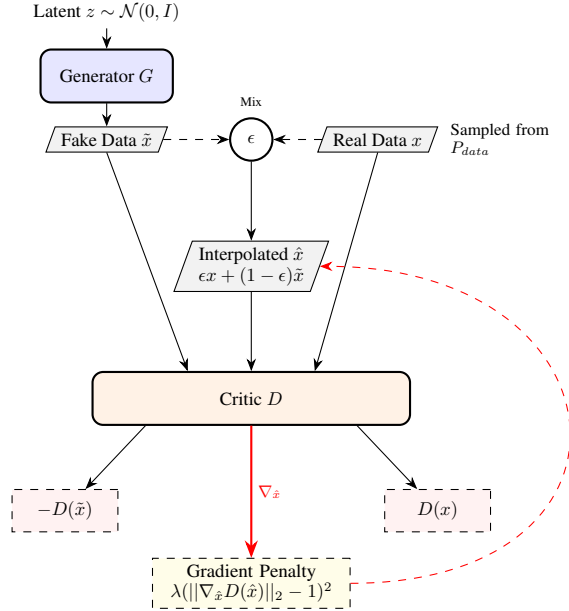


Figure 1: Overview of the WGAN-GP architecture.

3.1.2 Training Objective

Let $x \sim p_{data}$ be real images and $z \sim \mathcal{N}(0, I)$ be latent noise. The generator $G_\phi(z)$ produces synthetic images. The critic $D_\psi(x)$ estimates Wasserstein distance. WGAN-GP optimizes:

$$\max_{\psi} \mathbb{E}_{x \sim p_{data}} [D_\psi(x)] - \mathbb{E}_z [D_\psi(G_\phi(z))] - \lambda \mathbb{E}_{\hat{x}} (\|\nabla_{\hat{x}} D_\psi(\hat{x})\|_2 - 1)^2, \quad (1)$$

$$\min_{\phi} - \mathbb{E}_z [D_\psi(G_\phi(z))], \quad (2)$$

where $\hat{x} = \epsilon x + (1 - \epsilon)G_\phi(z)$ and $\epsilon \sim \mathcal{U}[0, 1]$.

3.1.3 Training Algorithm

We alternately update the critic and generator. The critic is trained for k steps with gradient penalty to enforce the 1-Lipschitz constraint, after which the generator is updated once. The procedure is summarized in Algorithm 1.

Algorithm 1 WGAN-GP Training

```
1: Initialize  $\phi, \psi$ 
2: for each training iteration do
3:   for  $k$  critic steps do
4:     Sample real batch  $x$ , noise batch  $z$ , and  $\epsilon \sim \mathcal{U}[0, 1]$ 
5:      $\tilde{x} \leftarrow G_\phi(z)$ ,  $\hat{x} \leftarrow \epsilon x + (1 - \epsilon)\tilde{x}$ 
6:      $L_D \leftarrow -\left(\mathbb{E}[D_\psi(x)] - \mathbb{E}[D_\psi(\tilde{x})]\right) + \lambda \mathbb{E}(\|\nabla_{\hat{x}} D_\psi(\hat{x})\|_2 - 1)^2$ 
7:     Update  $\psi \leftarrow \psi - \eta \nabla_\psi L_D$ 
8:   end for
9:   Sample noise  $z$ ,  $L_G \leftarrow -\mathbb{E}[D_\psi(G_\phi(z))]$ 
10:  Update  $\phi \leftarrow \phi - \eta \nabla_\phi L_G$ 
11: end for
```

3.2 Mini-DDPM

3.2.1 Model Architecture

Mini-DDPM parameterizes the noise predictor $\epsilon_\theta(x_t, t)$ with a lightweight U-Net. Given a noisy grayscale image x_t and timestep t , the network outputs a predicted noise map $\hat{\epsilon}_\theta(x_t, t)$. We use sinusoidal timestep embeddings followed by a small MLP and inject the embedding into each encoder and decoder block via additive conditioning. To reduce training cost, the network adopts a shallow design and omits self-attention at 64×64 (Fig. 2).

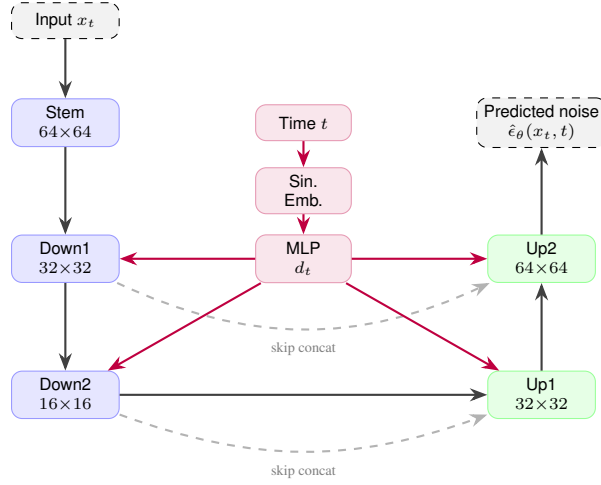


Figure 2: Mini-DDPM U-Net architecture.

3.2.2 Training Objective

We adopt the standard DDPM forward noising process:

$$q(x_t|x_{t-1}) = \mathcal{N}\left(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t I\right), \quad t = 1, \dots, T, \quad (3)$$

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \quad \bar{\alpha}_t = \prod_{i=1}^t (1 - \beta_i). \quad (4)$$

The model is trained to predict the injected noise via

$$\mathcal{L}_{\text{diff}}(\theta) = \mathbb{E}_{x_0 \sim p_{\text{data}}, \epsilon \sim \mathcal{N}(0, I), t \sim \mathcal{U}\{1, \dots, T\}} [\|\epsilon - \epsilon_\theta(x_t, t)\|_1]. \quad (5)$$

3.2.3 Training and Sampling Algorithms

During training, a timestep t is sampled for each clean image x_0 to construct a noisy observation x_t . The U-Net receives (x_t, t) as input and predicts the injected noise. Parameters are optimized by minimizing the discrepancy between predicted and ground-truth noise, as summarized in Algorithm 2.

Algorithm 2 Mini-DDPM Training (Noise Prediction)

- 1: Sample $x_0 \sim p_{\text{data}}$, timestep $t \sim \mathcal{U}\{1, \dots, T\}$, and noise $\epsilon \sim \mathcal{N}(0, I)$
 - 2: Construct $x_t = \sqrt{\alpha_t}x_0 + \sqrt{1 - \alpha_t}\epsilon$
 - 3: Predict $\hat{\epsilon} = \epsilon_\theta(x_t, t)$
 - 4: Update θ by minimizing $\|\epsilon - \hat{\epsilon}\|_1$
-

At inference time, generation starts from Gaussian noise $x_T \sim \mathcal{N}(0, I)$ and iteratively applies the learned reverse transition until a synthetic image x_0 is obtained. This iterative refinement improves structural fidelity but makes DDPM sampling slower than GAN-based generation. The sampling procedure is summarized in Algorithm 3.

When using noise prediction, the DDPM reverse mean can be written as

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \alpha_t}} \epsilon_\theta(x_t, t) \right), \quad \alpha_t = 1 - \beta_t. \quad (6)$$

Algorithm 3 Mini-DDPM Sampling (DDPM Reverse Process)

- 1: Initialize $x_T \sim \mathcal{N}(0, I)$
 - 2: **for** $t = T, T - 1, \dots, 1$ **do**
 - 3: $\hat{\epsilon} \leftarrow \epsilon_\theta(x_t, t)$
 - 4: $\mu \leftarrow \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \alpha_t}} \hat{\epsilon} \right)$
 - 5: Sample $z \sim \mathcal{N}(0, I)$ if $t > 1$, else set $z = 0$
 - 6: $x_{t-1} \leftarrow \mu + \sigma_t z$
 - 7: **end for**
 - 8: Return x_0
-

In our implementation, sampling uses the full $T = 1000$ denoising steps; faster schedules such as DDIM are not explored here.

4 Experiment

4.1 Dataset and Preprocessing

We use the Kaggle Chest X-Ray Images (Pneumonia) dataset [11]. Images are resized to 64×64 for generation experiments, converted to grayscale, and normalized to $[-1, 1]$. The official train/test split is preserved to avoid data leakage.

4.2 Synthesis of Pneumonia Chest X-ray Images

4.2.1 Experimental Settings

All generative models were trained using pneumonia images only. Each image was converted to grayscale, resized to 64×64 , transformed into a tensor, and normalized to $[-1, 1]$. We compared WGAN-GP and Mini-DDPM, where WGAN-GP uses a DCGAN-style generator and convolutional critic, and Mini-DDPM uses a lightweight U-Net with sinusoidal timestep embeddings.

Because the task is effectively single-class, Inception Score is not informative and is omitted. We therefore use FID between real and generated images,

$$\text{FID} = \|\mu_r - \mu_g\|_2^2 + \text{Tr} \left(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2} \right), \quad (7)$$

where (μ_r, Σ_r) and (μ_g, Σ_g) are the Inception feature statistics of the real and generated sets. Since the Inception encoder is pretrained on ImageNet rather than radiographic data, FID is used only as a reference metric, while downstream classification and qualitative inspection are emphasized.

For both models, FID was evaluated during the last five epochs, and the checkpoint with the lowest FID was used for final image synthesis. The main hyperparameters are summarized in Table 1.

Table 1: Main hyperparameters of the generative models.

Hyperparameter	WGAN-GP	Mini-DDPM
Input size	64×64	64×64
Channels	1	1
Normalization	$[-1, 1]$	$[-1, 1]$
Epochs	100	100
Batch size	32	32
Optimizer	Adam	Adam
Learning rate	1×10^{-4}	2×10^{-4}
Latent dimension	128	—
n_{critic}	5	—
λ_{gp}	10.0	—
Generator/Critic width	$ngf = 64, ndf = 64$	—
U-Net channels	—	$(64, 128, 256) / (256, 128, 64)$
Time embedding dim	—	32
Diffusion steps	—	1000
Beta schedule	—	linear, $10^{-4} \rightarrow 0.02$
Selection criterion	best FID in last 5 epochs	best FID in last 5 epochs
Generated images	3875	3875

4.2.2 Results

Convergence Analysis of the W-distance Curve:

As shown in Fig. 3, the Wasserstein distance fluctuates noticeably in the early stage of WGAN-GP training, then gradually stabilizes as training proceeds. This suggests that the critic first learns to separate real and generated distributions quickly, and later reaches a more stable balance with the generator. The overall downward-and-flattening trend indicates that the generated distribution becomes progressively closer to the real pneumonia distribution during training. Overall, the curve supports that adversarial training remains stable under gradient penalty, without obvious signs of divergence or collapse in our low-resolution setting.

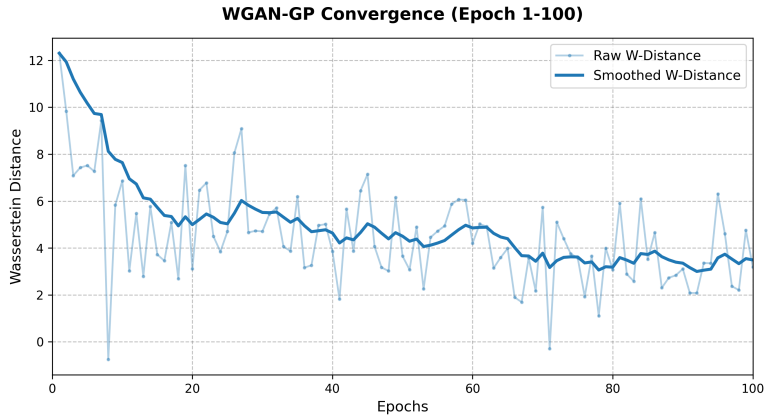


Figure 3: Wasserstein distance curve during WGAN-GP training.

Quantitative Comparison of Generated Image Quality and FID:

Figure 4 compares real images with samples generated by the two models at their best epochs.

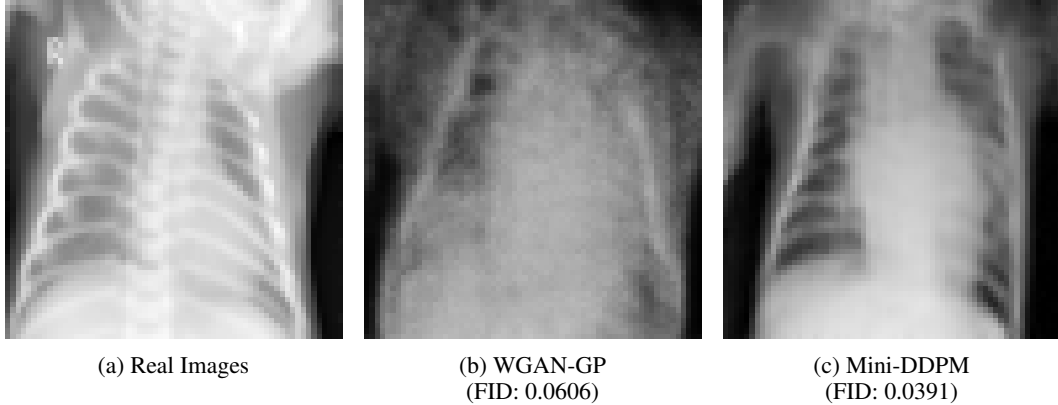


Figure 4: Visual comparison of real and generated samples.

The quantitative results demonstrate that the Mini-DDPM model achieved a superior FID score of **0.0391**, slightly outperforming the best score of the WGAN-GP model, **0.0606**. Combined with the visual qualitative comparison, it can be observed that while WGAN-GP is capable of generating images with relatively high fidelity, the images generated by Mini-DDPM excel in detail richness, noise suppression, and overall distribution alignment. This indicates that diffusion models, through their iterative denoising process, can capture and reconstruct the underlying manifold structure of real data more accurately than generative adversarial networks.

4.3 Downstream Task: ResNet-18 Pneumonia Diagnosis

4.3.1 Experimental Settings

To evaluate the usefulness of the generated images, we trained a ResNet-18 binary classifier (Normal vs. Pneumonia) under two settings: ImageNet-pretrained initialization and training from scratch. The goal is to test whether replacing part of the real pneumonia training set with synthetic images can preserve or improve downstream diagnostic performance.

All training sets were balanced to contain 1341 Normal and 1341 Pneumonia images. We compared three settings:

- **Baseline (Real Only):** 1341 real Normal images and 1341 real Pneumonia images randomly sampled from the original training set.
- **WGAN-GP:** 1341 real Normal images; the Pneumonia class is a 1:1 mixture of 671 real images and 670 synthetic images generated by WGAN-GP.
- **Mini-DDPM:** the same 1341 real Normal images and the identical subset of 671 real Pneumonia images used in the WGAN-GP group, plus 670 synthetic images generated by Mini-DDPM.

By fixing the 671 real Pneumonia images across the two generative groups, we isolate synthetic data as the sole independent variable. The dataset composition is summarized in Table 2.

Table 2: Dataset composition for downstream classification.

Dataset Setup	Normal (Real)	Pneumonia (Real)	Pneumonia (Fake)	Total Size
Baseline	1341	1341	0	2682
WGAN-GP (1:1)	1341	671	670	2682
Mini-DDPM (1:1)	1341	671	670	2682

Except for epochs and initialization, all other classifier hyperparameters were kept fixed to isolate the augmentation effect. The settings are summarized in Table 3.

Table 3: Hyperparameters for ResNet-18 training.

Hyperparameter	Value / Setting
Model Architecture	ResNet-18 (modified FC layer for binary classification)
Input Image Size	224×224
Channel conversion	Grayscale to 3 channels
Optimizer	Adam
Learning Rate	1×10^{-4}
Loss Function	Cross-Entropy Loss
Batch Size	32
Maximum Epochs	15 (ResNet18_Weights.IMAGENET1K_V1) / 50 (None)
Data augmentation	Random horizontal flip
Normalization (Mean)	[0.485, 0.456, 0.406]
Normalization (Std)	[0.229, 0.224, 0.225]

The generators were trained at 64×64 , and all images were resized to 224×224 only for downstream ResNet-18 evaluation. This design improves efficiency and stability: WGAN-GP is more prone to mode collapse at higher resolutions, while Mini-DDPM becomes substantially more expensive under limited hardware.

4.3.2 Results

The downstream classification task provides a practical test of whether the generated pneumonia images preserve diagnostically useful information. Overall, replacing half of the real pneumonia training samples with synthetic images does not degrade downstream performance, and in several cases yields small gains in AUC, specificity, or test accuracy.

Both settings also show some degree of overfitting. In the pretrained case, overfitting appears earlier because ImageNet initialization gives the classifier a much stronger starting point. In the training-from-scratch case, overfitting is also visible because the effective visual information is still limited by the original 64×64 generation process: resizing to 224×224 does not introduce new pathological details, while ResNet-18 still has enough capacity to fit this binary task closely. We therefore report the **best-performing epoch** rather than the final epoch for each setting.

ImageNet-pretrained ResNet-18:

We first evaluated the downstream task using ImageNet-pretrained ResNet-18, since transfer learning is the more standard and stronger baseline for medical image classification. The pretrained setting shows the clearest overfitting. In Fig. 5(a), the training loss drops very quickly within only a few epochs, which is expected because ImageNet-pretrained weights already provide strong low-level and mid-level features. However, Fig. 5(b) shows that validation accuracy does not improve monotonically with this continued reduction in training loss; instead, it stays within a relatively narrow range and fluctuates across epochs. This indicates that later training mainly fits the training set more closely rather than improving generalization. Again, this is why the comparison should be based on the best epoch rather than the last epoch.

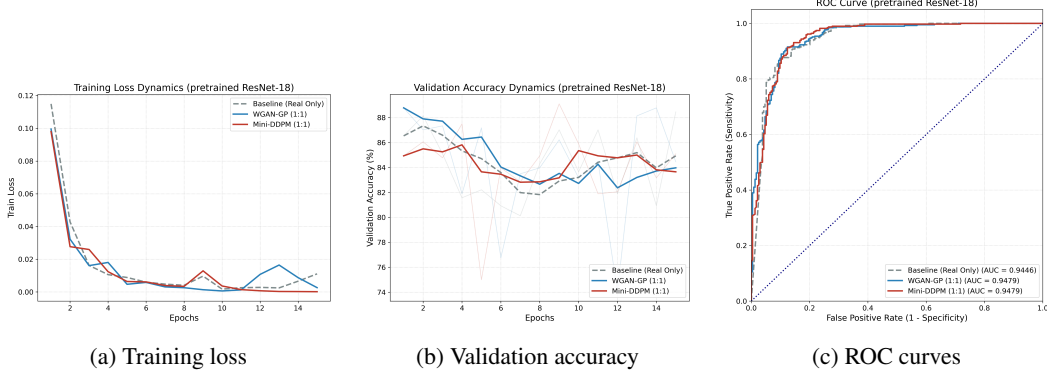


Figure 5: Downstream classification results of ImageNet-pretrained ResNet-18. (a) Training loss. (b) Validation accuracy. (c) ROC curves.

As reported in Table 4, Mini-DDPM achieves the best overall threshold-dependent performance, including the highest test accuracy (**89.10%**), the highest sensitivity (**98.21%**), and the best F1-score (**91.85%**). WGAN-GP again achieves the highest specificity (**78.21%**). In addition, both WGAN-GP and Mini-DDPM reach the best AUC (**0.9479**), slightly exceeding the real-only baseline (0.9446), which is also visible in Fig. 5(c). Therefore, even in a stronger transfer-learning setting that is more prone to overfitting, synthetic augmentation remains competitive and can slightly improve the final diagnostic performance.

Table 4: Classification results of ImageNet-pretrained ResNet-18.

Model	Best Epoch	Accuracy ($\uparrow$)	Sensitivity	Specificity ($\uparrow$)	F1	AUC
Baseline (Real Only)	15	88.46%	97.95%	72.65%	91.39%	0.9446
WGAN-GP (1:1)	1	88.78%	95.13%	78.21%	91.38%	0.9479
Mini-DDPM (1:1)	9	89.10%	98.21%	73.93%	91.85%	0.9479

ResNet-18 Trained From Scratch:

To verify that the observed benefit does not depend entirely on pretrained initialization, we further evaluated the same downstream task with ResNet-18 trained from scratch. As shown in Fig. 6(a), all three settings still drive the training loss to a very low level, but the validation accuracy in Fig. 6(b) does not continue improving accordingly. Instead, it fluctuates after the early epochs, again indicating overfitting. This effect is milder than in the pretrained setting, but it is still present. Therefore, the most meaningful comparison is again based on the best validation epoch rather than the final epoch.

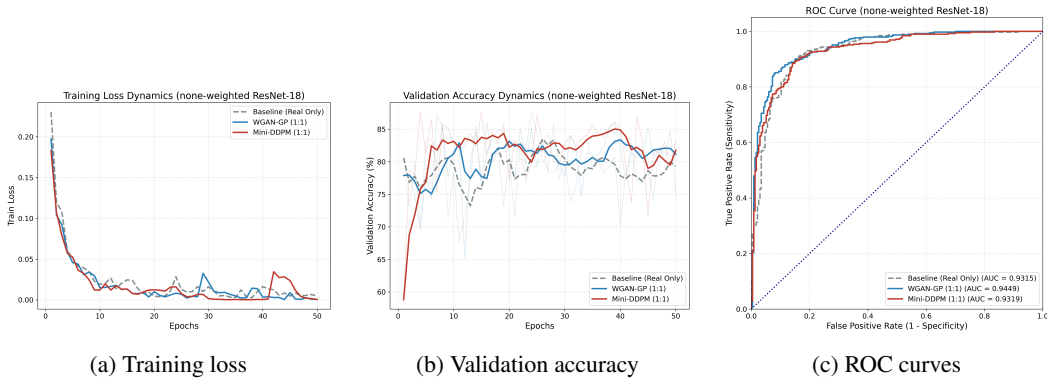


Figure 6: Downstream classification results of ResNet-18 trained from scratch. (a) Training loss. (b) Validation accuracy. (c) ROC curves.

Quantitatively, Table 5 shows that Mini-DDPM achieves the highest test accuracy (**87.66%**), while WGAN-GP attains the highest specificity (**80.34%**) and the best AUC (**0.9449**). The real-only baseline reaches the highest sensitivity (**94.10%**), but it also has the lowest specificity (**76.50%**), suggesting a stronger tendency to predict the pneumonia class. Consistent with this, the ROC curves in Fig. 6(c) show that WGAN-GP provides the best threshold-independent separability, while Mini-DDPM offers the best overall test accuracy. These results suggest that, even under random initialization, replacing half of the pneumonia class with synthetic samples does not hurt downstream performance and can improve the balance between sensitivity and specificity.

Table 5: Classification results of ResNet-18 trained from scratch.

Model	Best Epoch	Accuracy ($\uparrow$)	Sensitivity	Specificity ($\uparrow$)	F1	AUC
Baseline (Real Only)	24	87.50%	94.10%	76.50%	90.39%	0.9315
WGAN-GP (1:1)	7	87.18%	91.28%	80.34%	89.90%	0.9449
Mini-DDPM (1:1)	4	87.66%	92.56%	79.49%	90.36%	0.9319

5 Conclusion

This work compares WGAN-GP and Mini-DDPM as lightweight generators for pneumonia chest X-ray augmentation. Both models can produce visually plausible samples, while Mini-DDPM achieves the better FID and stronger visual fidelity. More importantly, replacing half of the real pneumonia training images with synthetic ones does not hurt downstream ResNet-18 classification and can slightly improve AUC, specificity, or accuracy.

Across the two generators, WGAN-GP tends to improve specificity and AUC, whereas Mini-DDPM provides the best overall balance, especially with pretrained ResNet-18. These findings suggest that the generated images retain clinically useful information and can serve as effective substitutes for real pneumonia samples in data-limited settings. At the same time, the observed downstream overfitting indicates that model evaluation should rely on the best-performing epoch rather than the final epoch. This also highlights that strong classifiers can easily overfit when the effective image information is limited by low-resolution synthesis.

This study also has several limitations. First, all generation experiments were conducted at 64×64 resolution, which improves efficiency but limits fine-grained radiographic detail. Second, FID was computed with an ImageNet-pretrained encoder and therefore serves only as a reference metric rather than a clinically grounded quality measure. Third, all downstream experiments were performed on a single public benchmark, so the generalizability of the observed gains to broader clinical settings remains to be validated.

Overall, our results support the practical value of lightweight generative augmentation under constrained computational resources. Future work should explore higher-resolution synthesis, more clinically meaningful evaluation metrics, and validation on broader datasets.

6 GitHub Link

Code repository: <https://github.com/TimXu1201/ECE-285>

References

- [1] Geert Litjens, Thijs Kooi, Babak Ehteshami Bejnordi, Arnaud Arindra Adiyoso Setio, Francesco Ciompi, Mohsen Ghafoorian, Jeroen AWM Van Der Laak, Bram Van Ginneken, and Clara I Sánchez. A survey on deep learning in medical image analysis. *Medical image analysis*, 42: 60–88, 2017.
- [2] Mateusz Buda, Atsuto Maki, and Maciej A Mazurowski. A systematic study of the class imbalance problem in convolutional neural networks. *Neural networks*, 106:249–259, 2018.
- [3] Luke Oakden-Rayner, Jared Dunnmon, Gustavo Carneiro, and Christopher Ré. Hidden stratification causes clinically meaningful failures in machine learning for medical imaging. *Proceedings of the ACM conference on health, inference, and learning*, pages 151–159, 2020.

- [4] Maayan Frid-Adar, Idit Diamant, Eyal Klang, Michal Amitai, Jacob Goldberger, and Hayit Greenspan. Gan-based synthetic medical image augmentation for increased cnn performance in liver lesion classification. *Neurocomputing*, 321:321–331, 2018.
- [5] Ali Madani, Mehdi Moradi, Alexandros Karargyris, and Tanveer Syeda-Mahmood. Chest x-ray generation and data augmentation for cardiovascular abnormality classification. In *Medical imaging 2018: Image processing*, volume 10574, pages 415–420. SPIE, 2018.
- [6] Ishaan Gulrajani, Faruk Ahmed, Martin Arjovsky, Vincent Dumoulin, and Aaron C Courville. Improved training of wasserstein gans. *Advances in neural information processing systems*, 30, 2017.
- [7] Walter HL Pinaya, Mark S Graham, Eric Kerfoot, Petru-Daniel Tudosiu, Jessica Dafflon, Virginia Fernandez, Pedro Sanchez, Julia Wolleb, Pedro F Da Costa, Ashay Patel, et al. Generative ai for medical imaging: extending the monai framework. *arXiv preprint arXiv:2307.15208*, 2023.
- [8] Mohammad Farukh Hashmi, Satyarth Katiyar, Avinash G Keskar, Neeraj Dhanraj Bokde, and Zong Woo Geem. Efficient pneumonia detection in chest xray images using deep transfer learning. *Diagnostics*, 10(6):417, 2020.
- [9] Sagar Kora Venu and Sridhar Ravula. Evaluation of deep convolutional generative adversarial networks for data augmentation of chest x-ray images. *Future Internet*, 13(1):8, 2020.
- [10] Daniel Schaudt, Christian Späte, Reinhold von Schwerin, Manfred Reichert, Marianne von Schwerin, Meinrad Beer, and Christopher Kloth. A critical assessment of generative models for synthetic data augmentation on limited pneumonia x-ray data. *Bioengineering*, 10(12):1421, 2023.
- [11] Daniel S Kermany, Michael Goldbaum, Wenjia Cai, Carolina CS Valentim, Huiying Liang, Sally L Baxter, Alex McKeown, Ge Yang, Xiaokang Wu, Fangbing Yan, et al. Identifying medical diagnoses and treatable diseases by image-based deep learning. *Cell*, 172(5):1122–1131, 2018. doi: 10.1016/j.cell.2018.02.010.